

Medical LLM Robustness

Speaker notes · n=50 final presentation · Group 2

SLIDE 1 OF 19

Title — Medical LLM Robustness

Hi, we're Group 2 — Medical LLM Robustness.

We tested whether a prompt-repair step actually helps medical QA models answer messy real-user questions, or whether it just looks like it helps.

This is the n=50 pilot: 11 answer models, 5 noise types, two repair pipelines (fixed_repair and self_repair). About 16 minutes of content.

→ *Move to the problem.*

Problem — Real users do not ask clean benchmark questions.

The motivation is simple. Real users don't ask clean benchmark questions.

MedQuAD is curated NIH/NLM content — careful, well-formed phrasing. Patients in the wild type at midnight: typos, vague wording, missing context, layperson terms.

In healthcare that isn't just a UX problem. Vague phrasing changes which medical concepts a model retrieves, which risks it surfaces, and how confident the user feels in the response.

So our research question: **Can a prompt-repair step recover the answer quality lost under noisy user input?**

The numbers up top: 11 models, 50 questions per model, 5 noise types — that's the scale we're working with.

→ *Before methodology, let's make sure we share vocabulary.*

Audience bridge — The project in one sentence

One-sentence version: we take real medical questions, make them messy like real-user input, try to repair them, then check whether the LLM answer gets better or worse.

The flow at the top is the pipeline: clean question → noisy version → repaired version → model answer → score vs reference.

Four terms to keep in mind:

- **Noise** — a controlled way of making the question harder (typo, vague phrasing, missing detail, layperson wording, overgeneralization).
- **Repair** — a rewrite step that tries to make the messy question clearer WITHOUT answering it.
- **Robustness** — does the model still answer well when input is messy?
- **Recovery** — does repair pull the answer back toward what we'd get on the clean question?

These four words cover everything in the rest of the talk.

→ *With shared vocabulary, here's the actual experimental design.*

Design — Two n=50 experiments separate fairness from realism.

Three pipelines per question:

- Clean → answer model → clean answer (our baseline).
- Noisy → answer model → noisy answer (the degradation).
- Noisy → repair → answer model → repaired answer (the recovery test).

We ran this structure in two modes, on purpose, because they answer different questions.

fixed_repair: GPT-5.4 generates ONE shared noisy set and ONE shared repaired set. All 11 models see identical inputs. Best for fairness — every model is judged on the same prompts.

self_repair: all models see the same noisy questions, but each model repairs its own. That's closer to a deployed system where the answer model also handles repair.

The two modes give us a controlled comparison and a realistic comparison.

→ *Here's exactly what was tested.*

Setup — What was tested

The scale: 11 answer models × 50 questions × 5 noise types = 550 rows per mode, evenly distributed by round-robin so every model sees 10 examples of each noise type.

Models span four families:

- Closed API: GPT-5.4.
- Open-weight instruction-tuned: Llama 3.1, Mistral 7B, Mixtral 8x7B, Qwen2.5 14B/32B, Qwen3 32B, Gemma 2 9B, Gemma 4 31B, Phi-3 med.
- Medical-domain: BioMistral-7B.
- Quantized local variants — 4-bit and Q6_K — to test whether quantization changes robustness behavior.

Noise types simulate the five most common failure modes we see in real user questions: typos and grammar slips, ambiguity, layperson rewording, incomplete prompts, and overgeneralization. Noise itself is generated by GPT-5.4-mini using fixed templates.

→ *How we evaluate is its own slide.*

Evaluation — Metrics used in the project

We use 11 metrics across 5 families. No single metric captures medical QA robustness, so we layer them.

- **Lexical overlap** — BLEU, chrF, ROUGE-L, Token F1, Exact Match. Each tolerant of different surface variations. Exact Match is mostly zero for free-form QA, included for completeness.
- **Semantic answer** — BERTScore. Catches paraphrase that lexical metrics miss.
- **Question intent** — embedding similarity between the clean question and the noisy/repared version. Tells us whether the user's actual ask was preserved.
- **Judge score** — G-Eval, where GPT-5.4-mini scores 1–5 on answer quality given question, reference, and answer.
- **Medical content** — coverage, precision, and F1 of medical terms vs the reference, using a MedQuAD-derived lexicon.

Don't memorize all of them — the next slide tells you which ones to actually watch.

→ *One reading rule before results.*

Evaluation — How to read the metrics

This slide is the most important interpretation rule in the talk.

Repair can clean up the question while still hurting the answer.

Intent metrics will improve almost by definition — repair's job is to clarify the user's ask, so similarity to the clean question goes up.

But a cleaner question doesn't guarantee a better answer. The model can still drop medical content, drift to a different topic, or lose specificity. We'll see all three failure modes in the evidence examples.

So we never rely on one metric. We pair intent with answer quality (G-Eval) and medical content (med_coverage / med_f1). When those agree, the story is clean. When they diverge, that divergence IS the story.

→ *With that said — here's the headline.*

Headline result — The n=50 pilot does not show systematic repair improvement.

The headline of the n=50 pilot: repair did not systematically improve answer quality.

Walk through the numbers: clean G-Eval averages 3.73. Noise drops it to 3.47. Repair pulls it back to **3.35 in fixed_repair** and **3.41 in self_repair** — both BELOW the noisy baseline.

Read that twice. On average, repair did NOT recover what noise took away. In fixed_repair it actually made things slightly worse on average.

(Pause for a beat — this is the counterintuitive moment.)

Important caveat: averages mask variance. Some models recovered strongly. Some got much worse. We'll show that distribution next.

And: this finding is the OPPOSITE of what intent metrics show — those improved everywhere. That gap between intent and answer quality is the actual finding.

→ *Let's open up the average per model.*

Model comparison — Which models recover G-Eval after repair?

Bars show repaired-minus-noisy G-Eval — positive means repair helped, negative means repair hurt. Top of the chart sorted to put the biggest self_repair winners first.

Two clear positive outliers in self_repair:

- **Qwen3 32B at +1.32** — biggest gain in the study.
- **Mixtral 8x7B at +0.66.**

Both of these models can rewrite their own bad input usefully. Their fixed_repair gain is much smaller, which suggests it's the SELF-rewriting capability that helps, not just having a cleaner prompt.

Most other models — Phi-3 med, Gemma 4 31B, Mistral 7B, Qwen2.5, Llama, GPT-5.4 — are flat or worse after repair. Several lose 0.3 to 0.6 G-Eval points.

Fixed-repair (the controlled mode) is even more conservative: only Qwen3 and BioMistral marginally improve.

Takeaway: **repair effectiveness is model-specific.** It is not a free lunch you can apply across the board.

→ *It's also not just about model — noise type matters.*

Noise breakdown — Repair behaves differently by noise type.

Same delta — repaired minus noisy — broken down by noise type instead of by model.

Self-repair helps slightly on **typos/grammar** and **incomplete** questions. These are surface-level issues; once the question is cleaned up, the model can answer.

Layperson phrasing and overgeneralization stay difficult — repair often locks in the wrong framing or broadens too far. We'll see exactly that in slide 13.

Practical implication: repair should be **CONDITIONAL**. Apply it to typos and incomplete prompts; don't apply it blindly to ambiguous or overgeneralized prompts where it might commit to the wrong interpretation.

→ *And here's the deeper pattern across the full metric stack.*

Metric tension — Repair cleans up intent more reliably than it improves answers.

This table is the core finding compressed into one view.

For each metric we count, across the 11 models: how many improved, how many got worse, how many stayed flat after repair.

The standout row is at the bottom: **Intent preservation — 11 / 0 / 0** in BOTH modes. Every single model's repaired question was closer to the clean question than its noisy version. Repair, in that narrow sense, always works.

Now look at every other row:

- G-Eval: 2 improved / 8 worse in fixed mode. 2/9 in self mode.
- BERTScore: 2/9 / 3/8.
- **med_coverage in self_repair: 0/11/0.** Every single model lost medical-term coverage.
- med_f1: 4/7 in fixed, 1/10 in self.

Core pattern: repair makes the question better, but not the answer. If we had only reported intent, we'd be telling a misleadingly positive story.

→ *Now let me show concrete examples of what's actually happening.*

Evidence example — Incomplete wording can make the answer collapse.

GPT-5.4, incomplete-noise question.

Clean question: "Do you have information about Air Pollution." Noisy: "do u have any info on bad air stuff and what it can do."

Look at the noisy answer: literal one-line fragment. "Yes — bad air." G-Eval crashes from 4 to 1. med_coverage from 27 to 2.

Repair turns the question into "What health effects can poor air quality or air pollution have on the body" — a clean medical ask. The answer comes back: irritation, asthma, COPD, lung function. G-Eval back to 4. med_coverage recovers to 16.7.

This is repair working as designed. Most of the lost medical content came back. This is the kind of case that makes you think repair is a safety layer.

→ *But it doesn't always work this cleanly.*

Evidence example — Noisy wording can shift the topic.

Same model, same noise type, different question — and the failure mode shows up.

Clean topic: "Anabolic Steroids." Noisy: "those muscle-building stuff." Repair rewrites it as "health risks of muscle-building supplements OR anabolic steroids."

Notice what just happened. Repair widened the question. The original clean topic was anabolic steroids specifically — a regulated, prescription-context medical concept. The repaired version now ALSO includes generic supplements like protein powder.

The answer drifts. The repaired answer talks mostly about protein powder, creatine, pre-workouts. The medical concept shifted before the answer model ever saw the question.

G-Eval is 3 — slightly better than the noisy 2 — but med_coverage went 28 → 7 → 13. We never recover the original medical specificity.

This is the central failure mode: question repair commits to the wrong interpretation. The answer model is now answering a different question.

→ *Layperson noise has the same trap.*

Evidence example — Layperson phrasing changes the frame.

Layperson noise. The clean question is just "What is abortion." The noisy version is a long patient-language paraphrase: "what does it mean when a pregnancy gets ended early or the baby is not kept..."

Repair handles this well. It restores a concise medical distinction: spontaneous abortion (miscarriage) versus induced abortion. The answer comes back tight and clinically accurate. G-Eval recovers to 4. med_coverage holds at 23.

The takeaway here is the contrast with the previous slide: layperson noise CAN be repaired well, but only when repair correctly identifies the medical category. When the layperson framing is ambiguous, repair has to guess — and sometimes guesses wrong.

→ *One more example, showing why we don't trust G-Eval alone.*

Evidence example — A better judge score can still lose medical content.

Mixtral 8x7B, typos-noise question on anal disorders. The judge G-Eval scored the repaired answer **5 — a perfect score**.

But look at med_coverage: **22.2** → **19.4** → **16.7**. Coverage falls steadily through both noise AND repair, even as G-Eval improves.

The repaired answer reads as fluent and well-organized, which is what GPT-5.4-mini judges on. But the medical content audit sees fewer specific terms — fewer named conditions, less precision.

This is exactly why we layer metrics. A judge alone would tell us the system improved. The medical lens says we lost specificity. Both are true. Whether that matters clinically depends on what's missing.

→ *So what does all this evidence add up to?*

Interpretation — What the n=50 evidence means

Three things from the n=50 evidence.

One: repair is useful but bounded. It can fix surface noise — typos, missing words, layperson terms. It cannot reliably recover medical meaning once the prompt has actually drifted in topic.

Two: question repair adds another failure mode. The repair model can commit to the wrong interpretation, broaden the target, or shift the medical category — all BEFORE the answer model ever sees the question. Slide 13 was the textbook case.

Three: metric choice changes the story. Intent metrics improve across the board. Judge scores sometimes improve. Medical content quietly falls. Multi-lens evaluation isn't optional in this domain.

If you want a one-line answer to the research question: **prompt repair is a conditional tool, not a safety layer.**

→ *We're being deliberately conservative about what this pilot can claim.*

Limitations — What this pilot cannot yet claim

What this pilot CANNOT yet claim, on purpose.

- **Pilot scale.** 50 questions per model is enough for signal-finding, not for confidence intervals on per-model effects.
- **Synthetic noise.** GPT-generated noise is consistent and controlled. Real patient language is messier, multimodal, and includes things like image attachments and EHR context.
- **Single dataset.** MedQuAD is curated NIH/NLM content. Patient portal messages, telehealth chats, and EHR notes look different and would likely show different repair behavior.
- **Judge model overlap.** G-Eval uses GPT-5.4-mini; our repair/noise stack also uses GPT-family components. Not fully independent — possible self-favoring bias.
- **Domain metric limits.** med_coverage catches lost terms but doesn't judge whether a missing term would matter clinically.

These aren't excuses — they're the boundaries of what the n=50 evidence can support. Per-model effects need scale, real noise needs real data, and clinical safety needs human review.

→ *Here's how we'd address those.*

Next steps — What would turn the pilot into a stronger study

Four things to turn this from pilot into a stronger study.

- **Scale to n=1000.** Same fixed_repair and self_repair structure with confidence intervals. Lets us make per-model claims with statistical support.
- **Validate repair outputs.** Reject rewrites that answer the question, ask for clarification, or inject facts not in the noisy input. Right now repair is a free-text rewrite — we should constrain it.
- **Human or clinical review.** Pair the automatic metrics with a harm-tier rubric for misleading, unsafe, or clinically dangerous answers. The judge can't catch what would actually hurt a patient.
- **Noise-specific policy.** Apply repair selectively. Typos and incomplete questions probably benefit. Overgeneralized or ambiguous prompts probably need a clarifying question instead of a rewrite.

The bottom line — and this is what we'd want a deployment team to take away:

Prompt repair should be evaluated as a CONDITIONAL tool, not assumed to be a universal safety layer.

→ *That's the talk.*

Thank you

The main answer in one sentence:

Repair improved question intent in every model, but did NOT systematically improve medical QA answer quality in the completed n=50 pilot.

Happy to take questions — and especially counter-evidence. If anyone has examples of patient-portal data, deployment experience, or domain feedback we should be paying attention to, we'd love to hear it.

Thank you.